

README_TableTask_NSTE_Analysis

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This document describes the Network Symbolic Transfer Entropy (NSTE) stage of the TableTask cardiac synchrony pipeline. It explains the analysis workflow, required inputs, the purpose of each script, the adaptations made from the original laboratory implementation provided by Jackie, the selected analysis parameters, and the expected outputs. The goal is to provide sufficient documentation for Veronica and future research assistants to understand, reproduce, and maintain this stage of the pipeline.

1. Pipeline Position

Scripts 01–04 prepare the ECG recordings, derive heart-rate time series, segment them into trial-level paired signals, and compute synchrony metrics. The NSTE stage is performed after Step 4 and operates only on the paired 4 Hz heart-rate files generated by the preprocessing pipeline.

2. Script

05_script_tabletask_nste_and_plots_v6_final_EG.py

Primary analysis + plotting script.

Main workflow:

- Locate all paired 4 Hz HR trial files from Step 3.
- Read TimeRelTrialSec, HR_A_BPM, and HR_B_BPM for each dyad-trial file.
- Merge available metadata and QC labels from:
 - Step 1 analysis units
 - Step 3 segmentation summary
 - Step 3a RR consistency summary
- Sort samples chronologically and identify contiguous valid paired-HR segments.
- Exclude missing, irregularly sampled, or constant segments from computation without bridging gaps.
- Standardize participant A and participant B HR signals within each valid continuous segment.
- Divide each segment into:
 - 30-second windows
 - 6-second steps
 - 80% overlap
- Determine the prediction delay for each window using the cross-correlation-based procedure adapted from Jackie's laboratory workflow.
- For each window, evaluate embedding delays of:
 - $\tau = 1\text{--}13$ samples
 - equivalent to 0.25–3.25 seconds at 4 Hz
- Reconstruct the signal state space using:
 - embedding dimension = 3
 - selected τ
 - window-specific prediction delay
- Convert the reconstructed HR patterns into ordinal symbolic sequences.

- Calculate raw Symbolic Transfer Entropy in both directions:
 - STE_YX: participant B $\rightarrow$ participant A
 - STE_XY: participant A $\rightarrow$ participant B
- Estimate shuffle bias separately for both directions using 20 internal source permutations.
- Calculate normalized, bias-corrected NSTE:
 - subtract the mean shuffled-source STE bias;
 - normalize by the corresponding target conditional entropy.
- Generate 400 outer surrogate datasets by unrestricted permutation of participant B's HR samples.
- Recalculate the full STE/NSTE workflow for every surrogate and every τ .
- Derive direction-specific surrogate statistics:
 - null mean
 - null standard deviation
 - Z score
 - one-sided Monte Carlo p-value
- Calculate directional asymmetry:
 - $B \rightarrow A - A \rightarrow B$
- Directly test the observed STE and NSTE asymmetry against their surrogate asymmetry distributions.
- Calculate two-sided asymmetry p-values using the finite-sample correction:
 - $(\text{number of equally or more extreme surrogates} + 1) / 401$
- Save the following for every valid window $\times \tau$ combination:
 - STE and NSTE in both directions
 - directional asymmetry
 - shuffle-bias estimates
 - surrogate statistics
 - p-values
 - effective symbol count
 - timing information
 - condition, trial, dyad, eligibility, and QC labels
- Save each completed trial result immediately in results_by_trial/.
- Update partial file manifests and failure logs after every completed trial so completed results are retained if execution is interrupted.
- After all files finish, combine trial outputs into the full NSTE result table.
- Generate one STE and one NSTE asymmetry plot for each dyad.
- Mark significant plot points using the direct asymmetry p-value with Bonferroni correction across the 13 displayed τ values.
- Generate a descriptive optimal- τ summary for FF and FE conditions.
- Export final:
 - combined CSV and Excel tables
 - per-trial files
 - file manifest
 - failure log
 - plot manifest
 - run summary
 - output README

- Retain all trials in the computation. QC and main-analysis eligibility variables are carried forward as labels and are not used automatically to exclude data.

Supporting NSTE Functions

The main NSTE analysis script (05_script_tabletask_nste_v4_final_EG.py) relies on several supporting Python modules that implement the underlying symbolic transfer entropy calculations. These files are adapted from the original laboratory NSTE implementation and are called automatically during the analysis. They should remain in the same directory as the main script and generally do not need to be run individually.

- **calculate_STE.py**: Computes Symbolic Transfer Entropy (STE) between two symbolic time series and provides the directional information transfer estimates used by the NSTE pipeline.
- **delayRecons.py**: Performs delay-coordinate reconstruction of the symbolic time series by constructing embedding vectors from past observations before transfer entropy estimation.
- **F_EstimateProb.py**: Estimates the joint and conditional probability distributions required for entropy and transfer entropy calculations.
- **F_Int2Symb.py**: Converts continuous-valued heart-rate signals into symbolic sequences that form the basis of the symbolic transfer entropy analysis.
- **F_Integer2prob.py**: Converts encoded symbolic states into probability distributions used during entropy estimation.
- **f_nste.py**: Implements the core Network Symbolic Transfer Entropy (NSTE) algorithm by combining symbolic encoding, state-space reconstruction, probability estimation, and transfer entropy calculations into the final directional NSTE estimates.
- **f_predictiontime.py**: Determines the prediction delay used during transfer entropy estimation by evaluating candidate prediction horizons, following the methodology implemented in the original laboratory pipeline.

These supporting modules implement the mathematical components of the NSTE algorithm, while the main analysis script manages data loading, trial processing, metadata integration, surrogate testing, output generation, and reproducible analysis of the TableTask dataset.

3. Adaptation from Jackie's Original Laboratory Scripts

The objective of the adaptation was to preserve the original NSTE methodology while making the implementation compatible with the TableTask project. The statistical estimation procedure was not intentionally redesigned. Most modifications concern automation, reproducibility, and integration with the existing preprocessing pipeline.

Component	Original script	TableTask adaptation
NSTE estimator	Preserved	Core computational procedure retained.

STE calculation	Preserved	Directional STE retained.
Symbolization	Preserved	Original symbolic representation retained.
Surrogate methodology	Preserved	Same surrogate-testing logic retained.
Parameter defaults	Preserved	Default analysis settings retained unless required for integration.
Input discovery	Manual	Automatically detects paired HR files from the Step 3 output directory.
Metadata	Minimal	Automatically merges analysis-unit and RR reliability metadata.
Directory structure	Project-specific	Uses standardized TableTask folder hierarchy.
Output naming	Project-specific	Standardized filenames and folder organization.
Logging	Limited	Run summaries, failure logs and manifests added for reproducibility.
Plot generation	Embedded/combined	Separated into a dedicated plotting script.

4. Parameters

The following parameters were retained from the original laboratory implementation unless noted otherwise.

Parameter	Value	Origin	Comment
Sampling frequency	4 Hz	TableTask	Determined by Step 3 interpolation.
Embedding dimension	3	Jackie	Adopted from original implementation. Publication-specific

optimization should
be confirmed by
Veronica.

τ search	1–13 samples	Jackie	Adopted directly.
Window lengths	10,15,30,45,60 s	Jackie	Adopted directly.
Window overlap	80%	Jackie	Adopted directly.
Internal shuffles	20	Jackie	Adopted directly.
Surrogates	400	Jackie	Adopted directly for final analysis.
Random seed	448	Added	Introduced only for reproducibility.

5. Inputs

Required input

- processed_ecg_hr/trial_segments_4hz/paired_trial_hr/
- Step 3 paired 4 Hz HR trial files (TimeRelTrialSec, HR_A_BPM, HR_B_BPM)

Optional metadata and QC inputs

- processed_ecg_hr/trial_segments_4hz/qc/03_trial_segmentation_summary.csv
- processed_ecg_hr/rr_reliability_qc/qc/3a_trial_dyad_rr_reliability.csv
- qc_outputs/veronica_analysis_units.csv

These optional files provide trial metadata, QC, reliability, and eligibility labels that are appended to the NSTE outputs. They are **not** used to automatically exclude trials from the analysis.

6. Outputs

Script 05 creates:

processed_ecg_hr/trial_nste_final_EG/

containing:

- Per-trial NSTE result files
- Combined window-level STE/NSTE results table
- File processing manifest
- Failure log
- Partial checkpoint manifests (updated during processing)
- Run summary
- Descriptive optimal tau summary
- Plot manifest
- NSTE directional asymmetry figures
- STE directional asymmetry figures
- Output README file

7. Notes

The adaptation was designed to preserve the statistical methodology used in the original laboratory implementation while making the workflow reproducible and compatible with the TableTask preprocessing pipeline. Any future methodological changes, including embedding dimension, surrogate count, window configuration, or inferential procedures, should be discussed with Veronica before they are adopted for a publication-quality analysis.